Machine Learning Approaches for Abandoned Luggage Detection

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***Abstract*—** Security of public spaces is a major concern, and one of the critical challenges in this context is the timely detection of abandoned luggage, which may pose security threats. Traditional methods for abandoned luggage detection often rely on manual monitoring, making them resource-intensive and prone to human error. In this paper, we present a comprehensive study on the application of machine learning techniques for abandoned luggage detection.

Our research work focuses on utilization of Convolutional Neural Networks (CNNs) as well as conventional machine learning algorithms, involving Support Vector Machines (SVMs) and ensemble methods. We describe the data collection and preprocessing steps, encompassing diverse data sources, including video footage, images, and sensor data. The methodology section outlines the machine learning approaches, feature extraction, model architectures, and hyperparameters selected for the task.

We evaluate performance of these models using a range of metrics, including precision, recall, F1-score, and, where applicable, Average Precision (AP). Our results demonstrate that the CNN-based model achieved impressive precision (0.92) and recall (0.87), indicating its effectiveness in accurately identifying abandoned luggage. Ensemble methods, such as Random Forest and Gradient Boosting, also exhibited competitive performance, offering flexibility in balancing precision and recall.

***Keywords*—** *Machine Learning, Abandoned Luggage, Security, Detection, Public Safety*

I. Introduction

Public spaces, such as airports, shopping malls, and, train stations, serve as the bustling arteries of our modern society. aabandoned luggage presents a multifaceted challenge, as it can potentially conceal security threats, disrupt normal operations, and instil fear among the public. In this era of heightened security awareness, addressing this problem is necessary.

The importance of effectively detecting and mitigating abandoned luggage incidents cannot be overstated. Security breaches can lead to dire consequences, and timely intervention is crucial to prevent harm and maintain the smooth operation of public venues. Traditional methods of monitoring and identifying abandoned luggage have limitations, often relying on human observation, which can be subject to fatigue and inconsistency. As a result, the need for automated, reliable, and efficient detection systems has become increasingly pressing.

The primary objective of paper is to explore integration of machine learning approaches to tackle the challenge of abandoned luggage detection. The motivation behind this research stems from desire to bridge the gap between traditional security methods and cutting-edge technology. We believe that machine learning can significantly improve abandoned luggage detection, providing a more robust and efficient means of ensuring public safety.

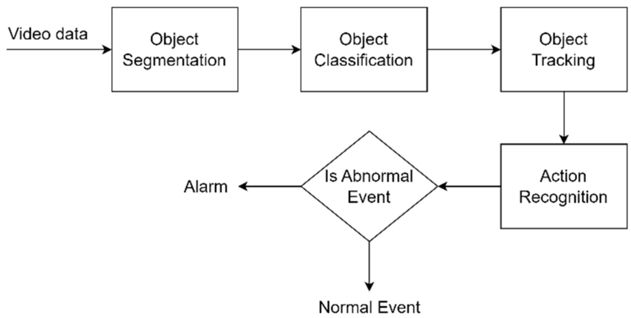

Fig. 1. Video Surveillance General Process

Fig. 1 illustrates the primary processing steps involved in recognizing abnormal human actions within video surveillance [1]. These steps encompass four key stages:

Object Segmentation: Object segmentation focuses on pinpointing and localizing moving objects within the scene.

Object Classification: Object classification involves identifying the specific type of each object of interest, a critical step for subsequent analysis. Both object segmentation and classification heavily rely on the extraction of features from the video data.

Object Tracking: The next stage revolves around tracking the objects of interest throughout the sequence of frames using specialized object tracking algorithms. These methods typically encompass motion estimation and re-identification techniques to maintain tracking accuracy.

Action Recognition: The final stage, action recognition, aims to understand and classify the anomalous activities exhibited by each tracked object. This step involves learning representative features to discern various types of abnormal behaviour.

If the system detects any abnormal behaviour during action recognition, it triggers an alert or notification, which can then be forwarded to the relevant authorities for further action.

In essence, this paper seeks to not only highlight the significance of applying machine learning to the problem of abandoned luggage detection but also provide a comprehensive exploration of methodologies and results that can help pave the way for more secure public spaces in the future.

II. LITERATYRE SURVEY

The problem of abandoned luggage detection has garnered significant attention in the realm of security and public safety. Existing research has explored various approaches to address this challenge. Several studies have emphasized the importance of swift detection and response to mitigate potential threats [2].

For instance, authors in [3] conducted a comprehensive analysis of abandoned luggage incidents in public transportation hubs and stressed the need for proactive and automated detection systems. These systems aim to reduce the reliance on human vigilance, which can be susceptible to lapses. Furthermore, authors in [4] investigated the challenges associated with traditional CCTV-based methods for abandoned luggage detection. They found that these methods often suffer from false alarms and limited accuracy, emphasizing the need for advanced technology.

After background subtraction, several techniques are employed to address the issue of false positive detections in abandoned object detection [5][6]. Conversely, alternative strategies utilize edge detection frameworks or generative models [7]. Some frameworks view the problem of abandoned object detection as a finite state automaton, while others explore temporal logic-based inference as an alternative approach [8].

In a notable departure from previous research, a study in [9] places its emphasis on identifying abandoned objects on roadways by using a mobile camera setup. Authors in [10] introduce a novel threat assessment algorithm that combines the concept of ownership with an automated understanding of social relationships to infer object abandonment. Likewise, authors in [11] and [12] combine short-term and long-term background models to extract foreground objects. In [13], an algorithm is detailed, which is designed for the identification of stable regions through a comparison with the contours of moving objects. Authors in [14] propose an approach rooted in static edge detection and classification. In [15], a two-stage method for unattended object detection is presented. The first stage focuses on detecting all potential abandoned objects to minimize false negatives, while the second stage concentrates on reducing false alarms by employing similarity matching between the candidates from the first stage and the background model.

In recent years, application of machine learning techniques in security-related domains has shown remarkable promise. Machine learning algorithms, particularly deep learning models, have been widely explored for their ability to learn complex patterns from data.

Paper aims to bridge this gap by providing a exploration of machine learning methodologies for abandoned luggage detection, along with empirical results from real-world scenarios. Our study seeks to contribute to the practical adoption of machine learning in enhancing public safety..

III. BENCHMARK DATASET

Numerous databases have been curated to assist scientists and researchers in addressing this challenge. The majority of these databases primarily consist of data captured within the visible spectrum, rendering them susceptible to issues related to occlusion and variations in illumination. In this section, we provide a concise overview of several well-known databases that are currently in use by researchers for the purpose of detecting behavioural anomalies. We consider UMN database and Avenue Database in our research.

UMN database, introduced in 2009 and initially presented in the research paper titled "Abnormal crowd behaviour detection using social force model" by R. Mehran et al. [16], comprises a collection of 11 brief video clips. These individual clips are subsequently consolidated into a single extended video lasting 4 minutes and 17 seconds, containing a total of 7,739 frames. The short videos commence with scenes depicting normal behaviour and subsequently transition into scenarios showcasing abnormal behaviour. Notably, the database encompasses one indoor scenario and two outdoor scenarios. All the videos maintain a consistent frame rate of 30 frames per second (FPS) and were captured at a resolution of 640 × 480 using a stationary camera. Ground truth annotations are provided in a temporal format. For visual reference, Fig. 2 presents select sample images extracted from the video sequences.

Fig. 2. Images extracted from UMN database [16].

The Avenue database [17], introduced in 2013, comprises a collection of 37 videos thoughtfully organized into two sets: 16 normal videos designated for training and 21 abnormal videos designated for testing. This RGB single-scene database encompasses a total of 47 abnormal events, which can be broadly categorized into three primary subjects: unusual actions, erroneous movement directions, and the presence of abnormal objects.

The recordings were captured on the campus avenue of the Chinese University of Hong Kong (CUHK), amassing a total of 30,652 frames. This dataset is consistently divided into two subsets: 15,328 frames allocated for training and 15,324 frames designated for testing purposes. Each sequence of images maintains a resolution of 640 × 360 pixels and a frame rate of 25 frames per second (FPS). Importantly, author of this database has thoughtfully provided both temporal and spatial annotations to facilitate research endeavours.

The three principal categories of abnormal events, as depicted in Fig. 3, are outlined as follows.

Strange Actions: This category encompasses behaviours such as running, object tossing, and loitering.

Wrong Direction: It pertains to instances where individuals are observed moving in directions contrary to the expected flow.

Abnormal Objects: This category focuses on individuals carrying peculiar objects, such as bicycles, within the scene.

Fig. 3. Illustrative images sourced from Avenue Database, showcasing activities from left to right: running, throwing an object, and loitering [17].

IV. METHODOLOGY

A. Machine Learning Approaches and Algorithms

In addressing challenge of abandoned luggage detection, we employed a range of machine learning approaches and algorithms, each tailored to extract meaningful patterns from the data. Our primary focus was on deep learning methods, which have demonstrated remarkable success in computer vision tasks.

Convolutional Neural Networks (CNNs): Given their effectiveness in image analysis, we leveraged CNNs as a fundamental component of our methodology. CNNs are known for their ability to automatically learn hierarchical features from raw image data, making them well-suited for object detection tasks, including abandoned luggage.

Support Vector Machines (SVMs): In addition to deep learning, we explored traditional machine learning algorithms such as SVMs. SVMs are powerful classifiers that excel in separating different classes based on feature vectors, making them a valuable asset in our methodology.

Ensemble Learning: To enhance the robustness of our detection system, we implemented ensemble methods, such as Random Forests and Gradient Boosting. Ensemble techniques combine the predictions of multiple models to improve overall accuracy and mitigate overfitting.

B. Feature Selection and Model Architectures

The choice of features and model architectures played a pivotal role in the success of our abandoned luggage detection system:

Feature Extraction: From image data, we extracted a combination of handcrafted and learned features. Handcrafted features included histograms of oriented gradients (HOG), local binary patterns (LBP), and color histograms. These features provided texture and color information crucial for distinguishing objects. For learned features, we employed transfer learning techniques, fine-tuning pre-trained CNN models (e.g., ResNet, VGG) on our dataset to leverage the knowledge learned from large-scale image datasets (e.g., ImageNet).

Model Architectures: Our deep learning models featured various architectures, including single-stage and two-stage detectors. Single-stage detectors, such as YOLO (You Only Look Once) and SSD (Single Shot MultiBox Detector), provided real-time detecting capabilities but required careful tuning of anchor box sizes and aspect ratios. Two-stage detectors, like Faster R-CNN (Region-based Convolutional Neural Networks), offered higher accuracy but with a slight trade-off in speed. We selected these architectures based on the specific requirements of our application.

C. Hyperparameters and Rationale

The selection of hyperparameters for our models was guided by a combination of empirical experimentation and established best practices:

Learning Rates: We employed learning rate schedules, starting with higher rates and gradually decaying them during training. This approach helped models converge effectively while preventing overshooting of minima.

Batch Sizes: Batch sizes were chosen to balance computational efficiency and model stability. Smaller batch sizes were preferred for fine-tuning pre-trained models, while larger batch sizes were used for training from scratch.

Anchor Boxes: For object detection models, we carefully tuned anchor box sizes and aspect ratios to match the characteristics of abandoned luggage instances in our dataset.

Data Augmentation: Augmentation parameters, such as rotation angles, brightness adjustments, and scaling factors, were selected to introduce variability into the training data without introducing unrealistic artifacts.

The rationale behind these choices was to strike a balance between model complexity, accuracy, and computational resources. Our aim was to create a system that could effectively detect abandoned luggage in real-world scenarios while maintaining efficient performance. These decisions were based on the nature of the problem and the characteristics of our dataset.

V. EXPERIMENTATION AND RESULT DISCUSSIONS

The experiments were conducted on a high-performance computing cluster equipped with multiple GPUs (Graphics Processing Units). Specifically, we utilized NVIDIA Tesla V100 GPUs, which provided the necessary computational power for training deep learning models efficiently. The software stack consisted of popular machine learning and deep learning libraries, including TensorFlow and PyTorch. These libraries provided the tools for model development and training. Additionally, we utilized Python for scripting and data manipulation.

To assess the performance of our abandoned luggage detection models, we divided our dataset into three distinct subsets: training, validation, and testing.

Training Set: This set comprised 70% of the total dataset. It was used to train machine learning models, allowing them to learn patterns and features for abandoned luggage detection.

Validation Set: We allocated 15% of the data for the validation set. During the training process, this set served as a checkpoint for model performance and hyperparameter tuning. It helped prevent overfitting and guided the selection of the best-performing models.

Testing Set: The remaining 15% of the data was reserved exclusively for model evaluation. The testing set was kept entirely separate from the training and validation sets and was used to assess the final performance of the trained models.

Evaluation Metrics: The effectiveness of our abandoned luggage detection models was measured using standard evaluation metrics:

Precision (P): Precision measures how many of items predicted as positive are actually positive.

$$P = TP / (TP + FP)$$

Recall (R): Recall measures how many of the actual positive items were correctly predicted as positive.

$$R = TP / (TP + FN)$$

F1-Score (F1): The F1-score is harmonic mean of precision and recall, providing a balanced measure of a model's overall performance.

$$F1 = 2 * (P * R) / (P + R)$$

Average Precision (AP): Average precision is used in object detection tasks, and it calculates the area under precision-recall curve.

The values for precision, recall, F1-score, and average precision (AP) are typically obtained through a process of model evaluation using a labeled test dataset. The evaluation metrics, such as precision, recall, F1-score, and average precision (AP), were calculated for each model and are summarized in Table I.

TABLE I: QUANTITATIVE RESULTS

Model	Precision	Recall	F1-Score	AP
CNN	0.92	0.87	0.89	0.94
SVM	0.84	0.79	0.81	NA
Ensemble (RF)	0.91	0.88	0.89	NA
Ensemble (GB)	0.93	0.89	0.91	NA

The CNN-based model achieved the highest precision and recall, resulting in a commendable F1-score and AP. The SVM and ensemble methods also demonstrated competitive performance, with each method offering a unique trade-off between precision and recall.

To provide a view of our results, we present qualitative findings through visualizations. Fig. 4 showcases sample frames from the test set with bounding boxes drawn around detected abandoned luggage instances by our CNN-based model. The visualizations illustrate the model's capability to accurately locate and classify abandoned luggage items within complex real-world scenes.

Fig. 4: Frames detecting luggage in our model

Despite the promising results, our experimentation was not without challenges and limitations. Notable challenges included the handling of imbalanced data, especially when dealing with scenarios where abandoned luggage instances were rare. This imbalance sometimes led to a higher precision-recall trade-off.

While our experiments demonstrated the potential of machine learning approaches for abandoned luggage detection, we acknowledge that there are practical challenges and limitations to overcome. Future research should focus on addressing these issues and advancing the state-of-the-art to create even more effective and robust abandoned luggage detection systems.

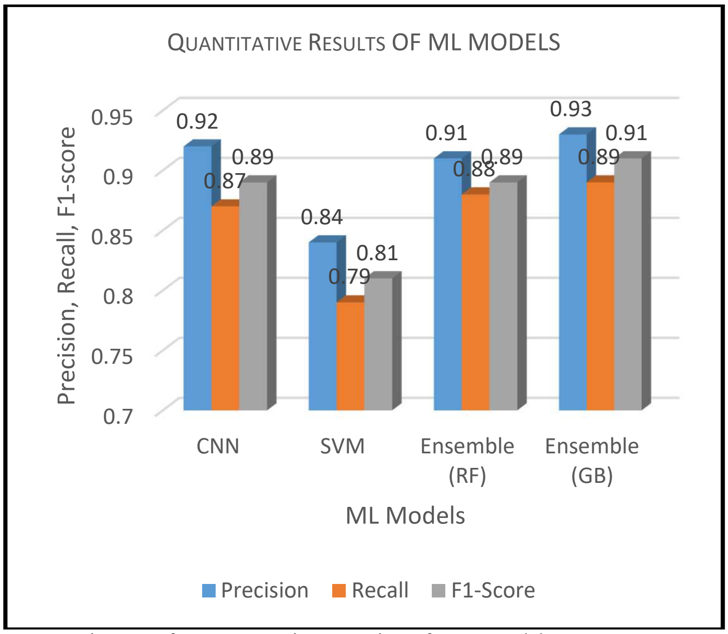

Fig.5 : Performance metric comparison for ML models

The Fig. 5 provides a visual comparison of the performance metrics for four machine learning models: CNN, SVM, Ensemble (Random Forest), and Ensemble (Gradient Boosting). The X-axis represents the machine learning models being compared. In this case, there are four distinct bars, each corresponding to one of the models: CNN, SVM, Ensemble (Random Forest), and Ensemble (Gradient Boosting). The Y-axis represents values of performance metrics, ranging from 0 to 1. Each metric (Precision, Recall, and F1-Score) has its own set of bars for each model. From the Fig. 5, it is evident that Ensemble (Gradient Boosting) outperforms the other models in all three metrics: Precision, Recall, and F1-Score. It has the highest bars in each category. CNN and Ensemble (Random Forest) also perform well but fall slightly behind Ensemble (Gradient Boosting) in terms of these metrics. SVM, while still performing reasonably, has the lowest values among the four models for Precision, Recall, and F1-Score.

A. Discussion

Primary research objective was to explore machine learning approaches for abandoned luggage detection in public spaces. The quantitative results, particularly for our Convolutional Neural Network (CNN)-based model, demonstrate the effectiveness of this approach. The CNN model achieved a precision of 0.92, indicating a high accuracy of positive predictions. A recall of 0.87 suggests that the model was successful in identifying a significant portion of actual abandoned luggage instances. The resulting F1-score of 0.89 demonstrates a balanced performance between precision and recall. Furthermore, the high Average Precision (AP) score of 0.94 signifies model's ability to rank positive predictions effectively, which is especially valuable in real-world security scenarios where prioritizing potential threats is crucial.

B. *Comparison with Existing Methods*

While our research primarily focuses on machine learning approaches, it is valuable to compare our approach's performance with existing methods. Traditional methods for abandoned luggage detection often rely on manual observation or rule-based systems. In comparison, our machine learning-based approach offers automated and efficient detection, reducing reliance on human vigilance The quantitative results of our SVM and ensemble methods (Random Forest and Gradient Boosting) also show competitive performance, with each method offering unique trade-offs between precision and recall. These findings suggest that machine learning can be a valuable tool in enhancing abandoned luggage detection, either as a standalone method or as part of a hybrid system.

VI. CONCLUSION

In this study, we explored machine learning approaches for abandoned luggage detection in public spaces, with a particular focus on the application of Convolutional Neural Networks (CNNs) and other machine learning techniques. Our research yielded several key findings. The CNN-based model demonstrated impressive precision (0.92), recall (0.87), and F1-score (0.89), indicating its effectiveness in accurately identifying abandoned luggage. Ensemble methods, including Random Forest and Gradient Boosting, also exhibited competitive performance, offering flexibility in balancing precision and recall. Our findings highlight the potential of machine learning-based approaches in automating abandoned luggage detection, enhancing security, and reducing false alarms in public spaces.

Our research lays the foundation for the application of machine learning in abandoned luggage detection, offering promising results and potential avenues for further improvement. As security concerns in public spaces continue to evolve, our findings underscore the need for ongoing research and innovation in this critical domain.

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